Day 9 — Speed Practice + Method Comparison

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve a mixed set of subtraction problems using both standard borrowing and Nikhilam, will compare the two methods on each problem, and will articulate a personal heuristic for when to pick which method.

Materials

- Comparison worksheet (10 problems, two-column answer space — one column per method)
- Stopwatches
- Whiteboard
- Final project work-time at the end

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mixed bases).
- Acknowledge: tomorrow is the summative.

Speed test (15 min) — head-to-head

Distribute the comparison worksheet. 10 problems, ordered roughly:

#	Problem	Likely faster method	Why
1	100 − 37	Nikhilam	Pure power-of-ten subtraction
2	87 − 24	Standard	Small, no borrow either way
3	1000 − 463	Nikhilam	Power-of-ten
4	532 − 178	Standard	No clean base nearby
5	10 000 − 5827	Nikhilam	Power-of-

#	Problem	Likely faster method	Why
			ten
6	7000 − 248	Nikhilam (with adjust)	Round multiple of 1000
7	9385 − 4716	Tie / standard	Mid-range subtrahend
8	50 000 − 12 345	Nikhilam (with adjust)	Round multiple of 10 000
9	8243 − 567	Nikhilam (with adjust)	Subtrahend close to 1000
10	4321 − 4118	Standard	Both small, similar magnitudes

Students solve each problem **once** by their chosen method, then again by the other method. Time both. Mark which method was faster.

Discussion (10 min) — When does each win?

Compile on board: which problems were faster with Nikhilam? Which with standard?

Heuristic that students will surface:

- **Nikhilam wins** when the minuend is a clean power of ten OR the subtrahend is close to a clean power of ten.
- **Standard wins** when both numbers are mid-range and the subtraction requires only one or two borrows.
- **Tie** when the numbers are small.

Write the heuristic on the board. Encourage students to copy it into their workbook.

Honest note: *“This is a real piece of mathematical maturity — knowing when to use which tool. Speed comes from knowing the choice, not from forcing one method everywhere.”*

Project work (12 min)

- Last in-class time before the showcase. Pairs work on their project artefact.
- Teacher circulates. Check: does each project have at least one Nikhilam-method example? Does each cite a source?

Wrap-up (3 min)

- Tomorrow's plan: 20-min summative quiz, then 3-min project presentations.
- Sūtra chant — final practice for tomorrow's recitation.

Student-facing content

When to use Nikhilam vs. Standard Borrowing

Situation	Faster method
Subtracting from 100, 1000, 10 000, …	**Nikhilam**
Subtracting a small number from a round multiple of 100/1000	**Nikhilam with adjust**
Both numbers in the middle, similar sizes	**Standard**
Both numbers small (under 100)	**Either**

The mature move: know both methods; pick the right one for the problem.

Homework

- Final project touches.
- 5 minutes of sūtra recitation practice — say it 5 times before bed.
- Bring all project materials tomorrow.

Differentiation

- **Tier up:** Mental challenge — do 10 mixed problems in your head with no paper. Time goal: under 3 minutes.
- **Tier down:** Worksheet only. Don't worry about choosing methods — solve each by Nikhilam where you can, standard otherwise.

Teacher notes

- The honest framing of "both methods are tools" matters. Avoid letting students walk away thinking "Nikhilam is always faster." It isn't. It's faster for *certain shapes* of problem.
- Some students will discover the heuristic naturally. Highlight their wording — let them name the rule.
- Project check is critical today. If any pair is behind, give them homework instructions explicitly.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.